A Machine Learning Approach to Home Run Classification in Baseball

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Outline

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Overview

The purpose of this project is to classify if a hit is a home run based on pitched and batted ball data.

Dataset-

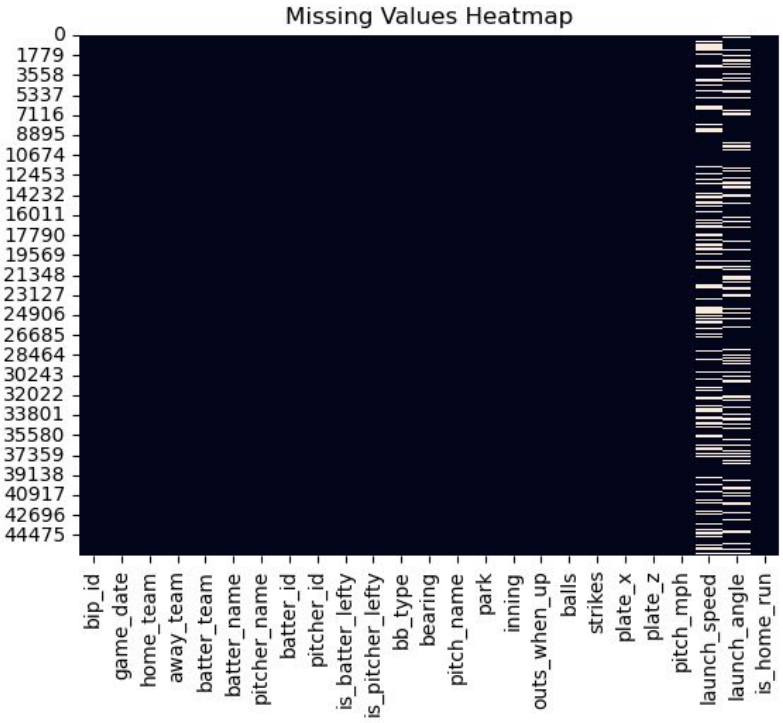
https://www.kaggle.com/datasets/jcraggy/baseball?utm_source=chatgpt.com

launch_speed	launch_angle	pitch_mph	plate_x	plate_z	balls	strikes	inning	outs_when_up	is_batter_lefty	is_pitcher_lefty	is_home_run
108.9	9.0	87.8	-0.15	1.97	1	2	1	0	0	1	0
75.0	49.0	86.5	-0.54	1.75	1	2	3	0	0	1	0
90.6	54.0	86.7	-0.42	3.15	0	2	1	0	1	1	0
66.7	27.0	86.2	-0.99	1.52	2	2	3	0	1	1	0
89.7	55.0	85.5	-0.44	2.33	1	1	4	1	0	1	0
99.3	37.0	96.0	-0.38	2.16	1	0	1	0	0	1	0
45.7	-47.0	95.0	-0.04	2.82	1	0	4	2	0	1	0
100.9	0.0	84.7	-0.09	1.21	1	2	6	0	0	1	0
105.4	28.0	89.5	-0.65	2.09	0	1	1	1	0	1	1
102.9	9.0	85.8	0.46	2.28	3	2	2	1	0	1	0
31.1	-33.0	94.9	1.68	3.86	2	2	3	1	1	1	0
46.6	-43.0	85.9	-0.14	2.35	0	0	5	1	1	1	0
101.7	38.0	92.1	-0.42	2.98	1	1	1	2	0	1	1
104.5	19.0	91.4	0.64	2.74	0	0	5	0	0	1	0
84.7	-3.0	87.5	-0.27	2.71	1	0	1	0	1	1	0
89.4	81.0	93.2	0.23	2.95	1	0	2	1	1	1	0
93.3	53.0	88.1	-0.18	2.42	1	0	1	1	1	1	0
96.5	63.0	94.0	1.03	2.85	3	0	2	2	0	1	0
104.5	24.0	94.0	-0.40	2.76	3	1	5	0	0	1	1
98.3	56.0	94.0	-0.34	2.04	3	1	3	0	0	1	0

Preprocessing

Clean Data

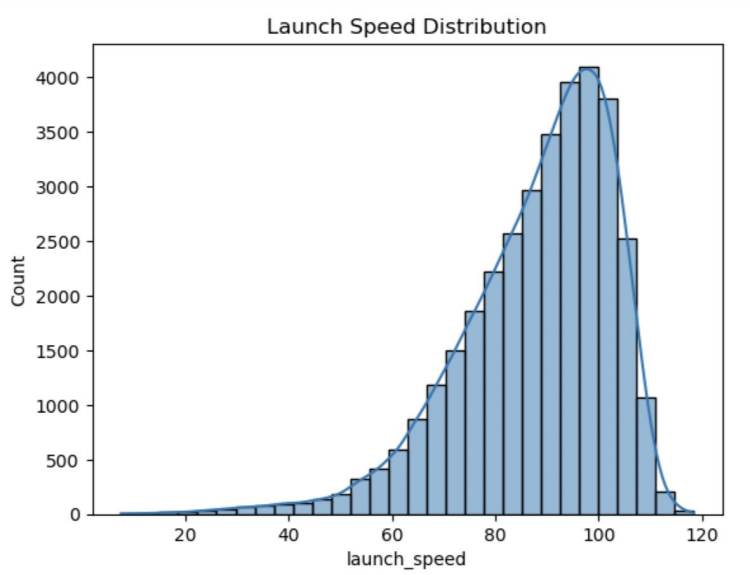
- Remove duplicate features
- Missing entries
- Remove features
- Encoded categorical features



Exploration

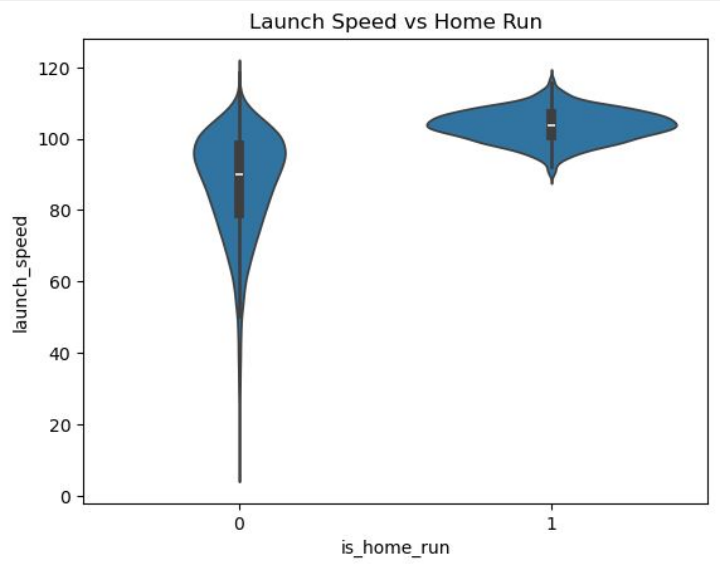
- Dataset: MLB batted-ball data (2020–2021)
- Around 46,000 observations, 25 variables

- Mix of:
- Numerical
- Categorical
- Target variable (is_home_run) is binary
- Only 5% of events are Home Runs.



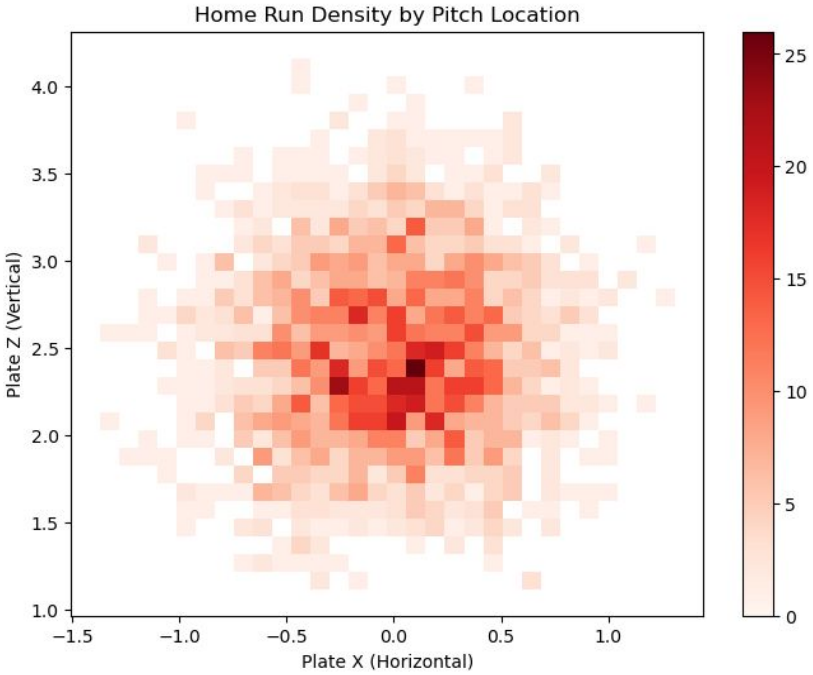
Exploration

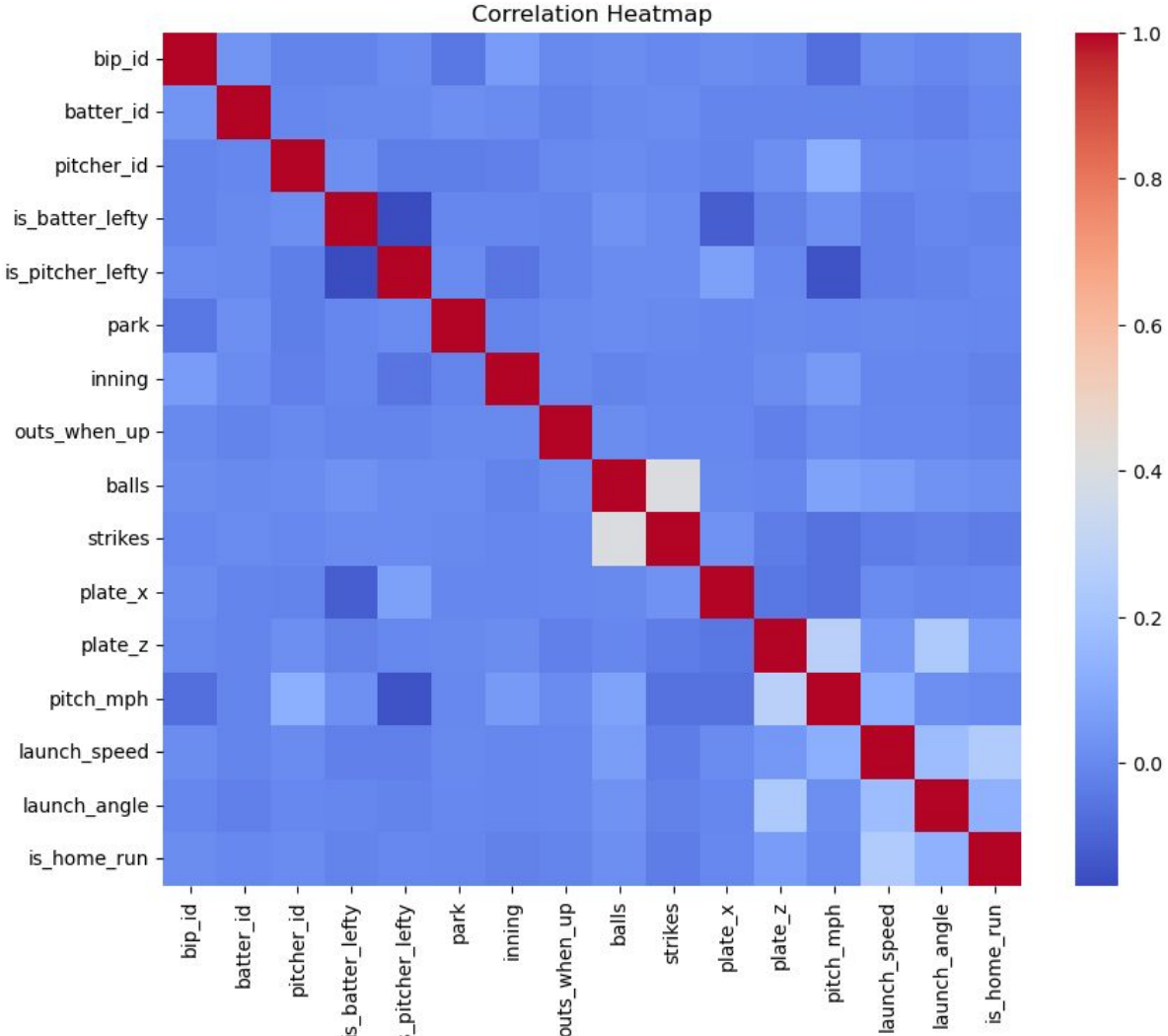
- **Key features: Launch speed Vs. Home Runs**
    - Launch speed is a very important predictor of home runs.
    - Most homeruns occur with a launch speed > 100 mph.
    - Some rare cases of high launch speed but no home run are due to other factors (launch angle).



Exploration

- **Non-linear patterns**
  - The data showed many non linear patterns in the correlation.
  - Pitch location is key when predicting home runs.
  - Home runs are common in pitches located in the center of the zone.



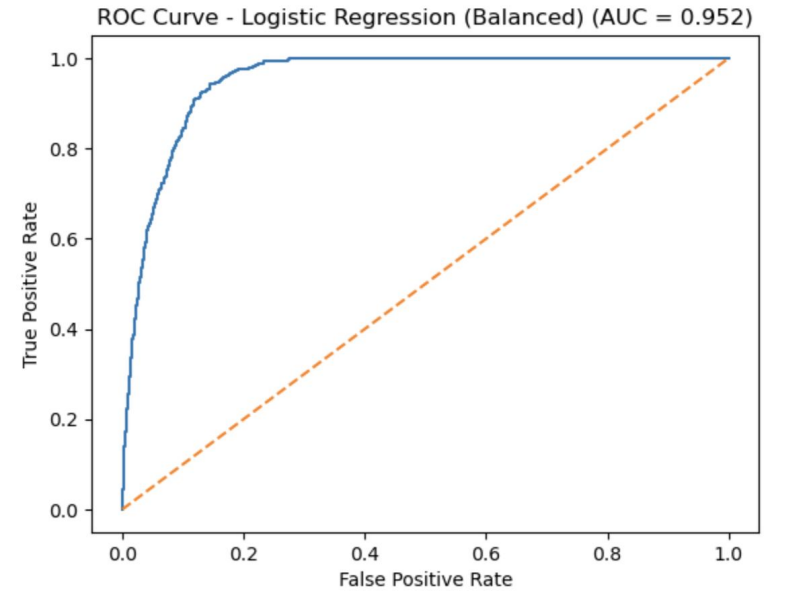

Correlation Heatmap
bip_id
batter_id
pitcher_id
is_batter_lefty
is_pitcher_lefty
park
inning
outs_when_up
balls
strikes
plate_x
plate_z
pitch_mph
launch_speed
launch_angle
is_home_run
1.0
0.8
0.6
0.4
0.2
0.0

Logistic Regression

Feature	Coefficient
Launch Speed	5.61
Launch Angle	2.77
Batted Ball Type (Fly Ball)	1.73
Batted Ball Type (Line Drive)	1.69
Home Team (NYY)	0.24
Away Team (ATL)	-0.21
Bearing Left	-0.28
Home Team (OAK)	-0.33
Batted Ball Type (Pop Up)	-1.71
Batted Ball Type (Ground Ball)	-2.07

Evaluation

- Accuracy= 0.87
- Precision= 0.29
- Recall= 0.91
- CV Average= 0.95
- F1-Score= 0.44



SVM

Parameter	Value
Kernel	rbf
C	1.0
Gamma	scale
Class Weight	balanced
Support Vectors	5897

SVM Model Comparisons

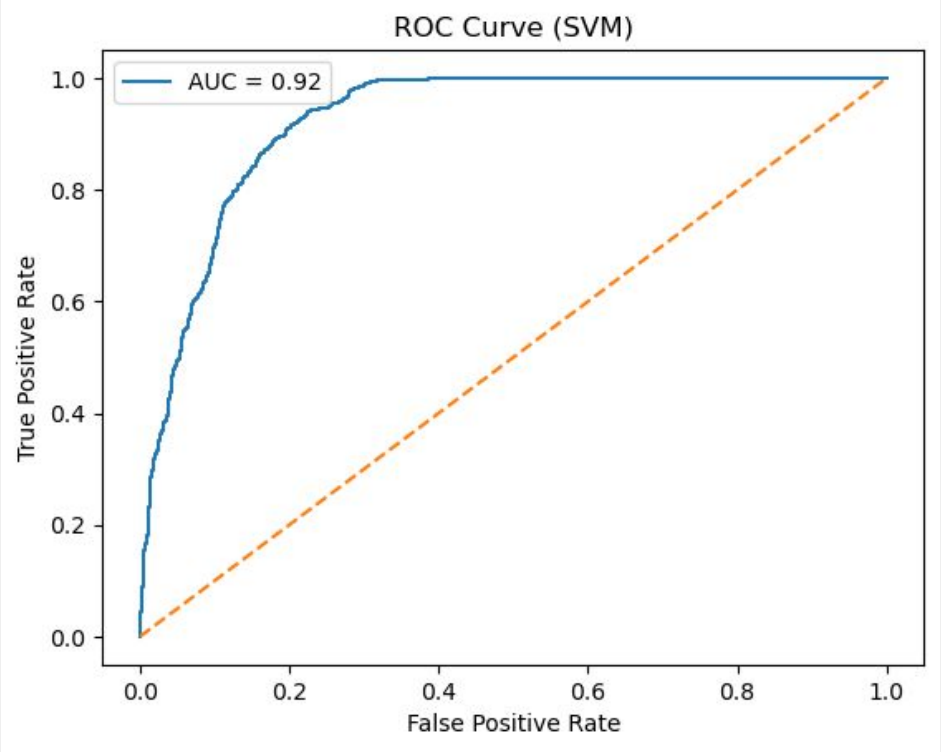
Base Model	
Accuracy	0.90
Precision	0.29
Recall	0.62
F1-Score	0.39

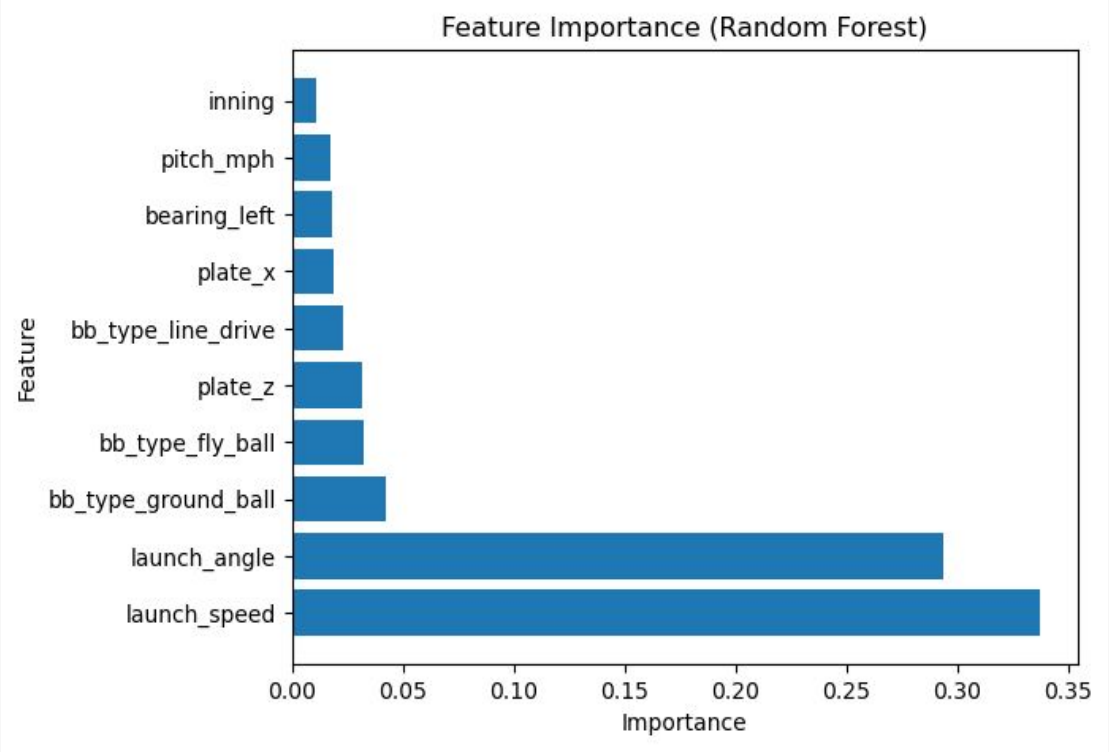
Unbalanced (RBF)	
Accuracy	0.95
Precision	0.00
Recall	0.00
F1-Score	0.00

Linear Kernel	
Accuracy	0.87
Precision	0.27
Recall	0.92
F1-Score	0.42

Evaluation

- Accuracy = 0.90
- Precision = 0.29
- Recall = 0.62
- F1-Score = 0.39





Random Forest

Random Forest Iterations

- GridSearchCV
  - max_depth, n_estimators, criterion, min_samples_split, max_features
- Feature Importance
  - top 2 features

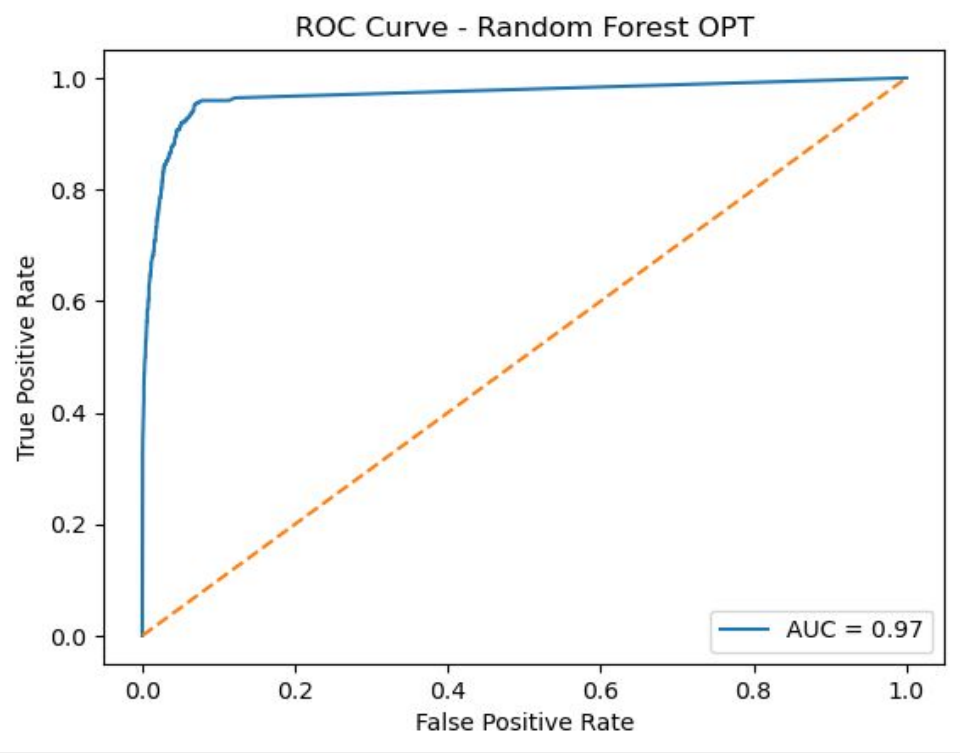
Base (Overfit Train)	
Accuracy	0.97
Precision	0.87
Recall	0.45
F1-Score	0.59

Param (max_depth=10, n_estimators=5)	
Accuracy	0.96
Precision	0.85
Recall	0.31
F1-Score	0.45

Min Features (“launch_speed”, “launch_angle”)	
Accuracy	0.97
Precision	0.76
Recall	0.67
F1-Score	0.71

Evaluation

- Accuracy = 0.97
- Precision = 0.76
- Recall = 0.67
- F1-Score = 0.71



Results

- Models have fairly consistent accuracy and AUC across
- Random Forest improved upon precision F1-Score
- Logistic Regression showed best recall
- “launch_angle”, “launch_speed” consistently top input features; with encoded “bb_type” features in top 10